PROJECT: HOSPITAL PATIENT FLOW – OPERATIONAL ANALYSIS

BUSINESS QUESTIONS SQL ANALYSIS

BY HEALTHCARE DATA ANALYSIS TEAM 3

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1. **HOW MANY PATIENTS ARE ADMITTED TO THE HOSPITAL EACH DAY?**

   This data shows the daily admission rate to the Hospital.
   The Hospital maintains steady admission volume daily, with numbers mostly ranging from 5 to 20, and occassional spikes of above 20 admissions. With a steady admission flow, the occasional spikes can put strain on the Hospital processes in areas of workload, bed spaces, and overall availability of resources.

2. **WHAT IS THE OVERALL AVERAGE LENGTH OF STAY (LOS) FOR ADMITTED PATIENTS?**

   The overall average length of stay is about 4days. This indicates a moderate efficiency in admission process. However, with the steady influx of patients and occasional spikes, this can impose constraints on the availability of bed spaces

3. **HOW DOES AVERAGE LENGTH OF STAY VARY BY DEPARTMENT?**

   Overall, the average length of stay appears to be consistent across all departments, with days ranging from 3.9 to 4.2days, suggesting a balanced patient flow and efficient admission process across all departments.

   Departments such as ICU, OBY/GN, and Medicine show shorter length of stay of about 3.9days, which could indicate efficient patient care and discharge process or considering departments like ICU, patients could be getting discharge prematurely. Overall though, this puts less constraints on availablity of bed spaces.

   However, with a slight variation, departments such as Orthopedics, surgery, and pediatrics shows an average length of stay of 4.1 days. This, although insignificant, may reflect that patients in these departments require extra care, probably due to disease complexities. It could also reflect inefficient discharge processes.

4. **HOW ARE ADMISSIONS DISTRIBUTED BY ADMISSION TYPE?**

   Admission volumes vary significantly across admission types.

   Emergency admission type records the highest rate with a record of 2755 admissions; accounting for 55% of the total admissions, this is followed by elective admission type accounting for about 30% of the total admission with a count of 1478 admissions, then followed by urgent admissions type accounting for only 15% with a count of 767.

   With this distribution, the hospital appears to be constantly responding to urgent, unpredictable cases. This places heavy demand on emergency departments, and requires round-the-clock staffing and rapid resource mobilization. Elective admissions, at 30%, show that planned procedures like surgeries or specialist treatments are still a significant part of operations, but they are secondary to emergencies. This balance suggests that while the hospital must maintain strong scheduling systems for elective care, it must also be flexible enough to reschedule or delay when emergencies surge. Urgent admissions at 15% highlight a smaller but still important category, often requiring quick intervention but not as immediate as emergencies.

Together, these proportions mean the hospital must prioritize emergency readiness, invest in triage efficiency, and ensure adequate bed capacity.

5. **WHICH ADMISSION TYPE IS ASSOCIATED WITH THE LONGEST AVERAGE LENGTH OF STAY?**

With an average length of approximately 4.3days, the urgent admission type accounts for the longest length of stay compared to other admission types. This could signify that patients admitted through urgent pathways may present with conditions that require more extended care than elective cases, but are not as immediately critical as emergency admissions. As a result, they may experience longer inpatient durations due to clinical complexity or delays in reaching discharge readiness.

From an operational perspective, this is significant because urgent admissions represent a category of patients that contribute the least to inflow, but occupy beds for longer periods. Their longer stays mean they can have a disproportionate impact on bed occupancy and patient flow.

6. **HOW DO DISCHARGE DISPOSITIONS BREAK DOWN ACROSS ALL ADMISSIONS? HINT: (*GROUP BY, COUNT, PERCENTAGE)***

The data shows that over 70% of patients are discharged home, meaning the hospital is generally successful in stabilizing patients enough for independent recovery. Around 16% go to rehabilitation centers, and about 10% are discharged to skilled nursing facilities, which highlights the need for structured recovery programs after hospital care. A small percentage, just under 3%, unfortunately, pass away during admission.

Operationally, this matters because it reflects the hospital's strengths and challenges. A high rate of home discharges suggests effective acute care and good patient outcomes, but it also means the hospital must have strong follow-up systems to ensure patients continue recovering safely. The significant share of rehab and skilled nursing discharges shows the importance of partnerships with external care providers and careful discharge planning to avoid gaps in care. The expired cases, though relatively few, emphasize the need for ongoing investment in critical care and palliative services. In short, the hospital's operations must balance immediate treatment with safe transitions, ensuring patients receive the right level of support after leaving its doors.

7. **DO CERTAIN DISCHARGE DISPOSITIONS CORRESPOND TO LONGER HOSPITAL STAYS?**

From the data, Patients who passed away (“Expired”) had the longest stays: 4.35 days on average. Those sent to rehab stayed 4.19 days. Patients discharged home stayed exactly 4 days. Those going to a skilled nursing facility had the shortest stays: 3.97 days.

Even though the differences are small, the pattern is consistent, and the more serious or final the outcome, the slightly longer the hospital stay. It also appears as though there is a standardized care process that drives the length of stay, regardless of the severity of diseases.

8. AT WHAT HOURS OF THE DAY DO MOST ADMISSIONS OCCUR?

At first glance, the numbers are relatively consistent, ranging from 187 to 259 admissions per hour, but the specific timing of these arrivals offers a clear roadmap for hospital management.

The busiest time for admissions is early in the morning, peaking significantly at 5 AM. Throughout the rest of the day, there is a steady flow of patients, with notable "mini-peaks" occurring around 10 AM and again in the evening at 8 PM (20:00). Conversely, the hospital sees its lowest volume of admissions during the late afternoon (around 4 PM to 5 PM) and the very early hours of the morning (between 2 AM and 4 AM). Understanding these "rush hours" is vital for keeping the hospital running smoothly. Because the highest volume occurs at 5 AM, It is important to ensure there are enough doctors, nurses, and intake clerks available during peak hours (5 AM and 10 AM) to prevent long wait times. The "slower" periods in the late afternoon can to clean rooms and process discharges so beds are ready for the evening and early morning influx.

9. ARE THERE SPECIFIC DAYS OF THE WEEK WITH CONSISTENTLY HIGHER ADMISSIONS*?(DOW: 0= SUNDAY, 6 = SATURDAY)*

Admissions throughout the week are relatively stable; although, the hospital experiences its highest volume of admissions on Sunday (737) and Friday (725). The middle of the week (Monday through Thursday) remains very consistent, with an average of about 710 admissions per day. Saturday seem is the quietest day of the week, dropping to 697 admissions.

These patterns are essential for effective hospital "forecasting." Since Sundays and Fridays are the peak days for admissions, the hospital must ensure that staffing levels, and resources are at full strenght over the weekend to handle the surge and maintain patient flow.

10. **WHICH DEPARTMENTS EXPERIENCE THE HIGHEST ADMISSION VOLUME?** ***(TOP 5 IS SUFFICIENT SO THAT THE SIGNAL ISN'T WEAKENED)***

There appears to be a fairly even distribution across all major departments, but Orthopedics (895**)** and Pediatrics (884**)** stand out as the highest-volume areas. Surgery, OB/GYN, and the ICU follow closely, each handling over 800 admissions. The Medicine department, while still busy, recorded the fewest admissions at 791.

Since Orthopedics and Pediatrics have the highest volume, these units likely require the most frequent supply restocks (such as casts, splints, or pediatric-specific equipment) and the highest number of specialized nursing staff.

11. **HOW DOES PATIENT AGE RELATE TO LENGTH OF STAY?** This data reveals that young adults (19 to 35) tend to stay the longest (4.26 days)**,** while middle-aged patients (51 to 65) have the shortest average stay (3.63 days)**.** Adults (30 to 50), elderly (66 and above), and pediatric (0 to 18) patients all fall in between, with averages around 4 days.

Operationally, this matters because hospital managers need to plan resources like bed availability, staff scheduling, and treatment costs based on how long patients typically remain admitted. Even small differences in average stay can add up across hundreds of patients, affecting efficiency and patient flow. For example, knowing that young adults stay slightly longer helps anticipate higher demand for beds in that group, while shorter stays among middle-aged patients may free up capacity more quickly. In short, these insights guide smarter planning to balance patient care with hospital resources.

12. **ARE OLDER PATIENTS MORE LIKELY TO BE DISCHARGED TO REHAB OR SKILLED NURSING?**

The data shows that among elderly patient (65 and above), more are discharged to rehabilitation facilities (237 patients) compared to skilled nursing facilities (168 patients)**.**

This means older patients are somewhat more likely to continue their recovery in a rehab setting rather than skilled nursing. Operationally, this matters because hospitals and care planners need to anticipate where patients will go after discharge. If rehab is the more common destination, hospitals must coordinate closely with rehabilitation centers to ensure enough capacity and smooth transitions. Skilled nursing facilities are still significant, but the higher rehab numbers highlight where demand is stronger, which can affect staffing, bed availability, and patient flow across the healthcare system.

13. **WHICH DIAGNOSES ARE MOST COMMONLY ASSOCIATED WITH HOSPITAL ADMISSIONS?**

The data shows the most common reasons patients are admitted to the hospital, based on diagnosis codes. The top conditions include A09 with 771 admissions**,** E11 with 737 admissions, and N39 with 734 admissions. Other frequent cases are I10, 708 admissions, O80, 703 admissions, J18, 680 admissions, and K35, 667 admissions.

Operationally, this matters because hospitals can use this information to plan resources and prioritize care. For example, knowing that infections and chronic conditions like diabetes are leading causes of admission helps guide staffing needs, supply management (such as antibiotics or diabetes care supplies), and preventive health programs. It also highlights where community health efforts could reduce hospital demand—like better infection control or diabetes management—ultimately improving patient outcomes and easing pressure on hospital capacity.

14. DO CERTAIN DIAGNOSES RESULT IN LONGER AVERAGE HOSPITAL STAYS?

The data shows the average length of hospital stay for different diagnoses. Patients admitted with diagnosis (N39) tend to stay the longest, averaging 4.17 days, while those with diagnosis (I10) have the shortest stays at about 3.9 days. Other conditions like diagnosis (A09), (E11), (K35), (O80), and (J18) all fall close to the 4-day mark.

Operationally, this matters because certain diagnoses require more hospital resources, extended monitoring and treatment, longer recovery times, which tie up beds and staff for longer periods. Understanding which conditions lead to longer stays helps hospitals plan capacity, allocate staff, and manage costs more effectively. It also highlights where preventive care or faster treatment pathways could reduce hospital burden and improve patient flow.

15. BASED ON YOUR ANALYSIS, WHERE DO YOU SEE THE BIGGEST PATIENT FLOW BOTTLENECK? (*THIS IS A WRITTEN ANSWER, NOT A SINGLE QUERY. DRAW ON YOUR FINDINGS FROM Q1–14*).

The biggest patient flow bottleneck appears to be centered around bed availability, driven by the combination of steady admissions, emergency case dominance, and a relatively fixed length of stay across patients. From the data, admissions remain consistently high on a daily basis, with occasional spikes and peak periods (especially early mornings and weekends), meaning the hospital is almost always operating near capacity. This pressure is further intensified by the fact that emergency admissions account for over half of all cases, making inflow highly unpredictable and difficult to control or schedule.

At the same time, the average length of stay remains stable at around 4 days across departments, diagnoses, and patient groups. While this suggests a standardized and somewhat efficient care process, it also indicates limited flexibility in how quickly patients can be discharged. In other words, bed turnover is relatively fixed, and cannot easily adjust to match sudden increases in admissions. This becomes more significant when considering urgent admissions, which, although fewer in number, have the longest average stays, thereby occupying beds for longer periods and slowing down overall patient flow.

Additionally, discharge patterns reveal another layer of constraint. A notable proportion of patients are discharged to rehabilitation and skilled nursing facilities, which introduces dependency on external systems. Delays in securing placement or coordinating transfers can result in patients remaining in the hospital even after they are medically fit for discharge, further reducing bed availability.

Overall, the bottleneck is not caused by a single factor, but by the interaction between high and unpredictable inflow, steady but inflexible patient outflow, and discharge delays. This imbalance limits the hospital's ability to create capacity quickly enough during peak periods, ultimately putting strain on bed spaces, staff workload, and overall resource utilization.